

EMPLOYEE ENGAGEMENT, SATISFACTION, AND BURNOUT DIAGNOSTIC ANALYSIS FOR PALO ALTO NETWORKS USING POWER BI

Project Report

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Technology Used

- Microsoft Power BI
- Power Query
- DAX (Data Analysis Expressions)
- Microsoft Excel

Project Domain

HR Analytics & Employee Experience Intelligence

Submitted to

Unified Mentor Private Limited.

ABSTRACT

This project focuses on analyzing employee engagement, satisfaction levels, burnout risk indicators, and work-life balance patterns at Palo Alto Networks using Business Intelligence and HR Analytics techniques.

The analysis integrates employee satisfaction dimensions, workload indicators, burnout signals, and engagement metrics to create a preventive employee well-being diagnostic framework.

The dashboard includes KPI cards, engagement index calculations, burnout risk scoring, workload stress analysis, role-level comparisons, tenure analysis, and manager intervention panels.

Interactive filters such as department selection, job role filtering, overtime analysis, and career-stage segmentation enable dynamic employee experience analysis.

The project demonstrates how Business Intelligence tools can transform employee data into actionable insights that support proactive HR interventions and organizational well-being strategies.

INTRODUCTION

Human Resource Analytics is the process of collecting and analyzing employee data to improve workforce management and organizational effectiveness.

Organizations increasingly recognize that employee engagement and well-being directly influence productivity, innovation, employee retention, workplace satisfaction and long-term organizational performance.

Palo Alto Networks operates in a highly demanding cybersecurity environment where employees may experience heavy workloads and burnout pressures.

Traditional attrition analysis identifies employee exits after they occur. However, this project shifts analysis toward preventive diagnostics by identifying burnout signals and engagement deterioration before attrition occurs.

The project develops an Employee Engagement and Burnout Diagnostic Dashboard using Power BI.

PROBLEM STATEMENT

Palo Alto Networks currently lacks:

- A unified employee engagement index
- Visibility into burnout-prone employee groups
- Understanding of work-life balance and overtime interactions
- Early identification of disengagement signals
- Preventive intervention mechanisms

As a result, HR decisions often become reactive rather than proactive.

OBJECTIVES

- To create an employee engagement index
- To identify burnout risk groups
- To analyze work-life balance conditions
- To evaluate workload stress indicators
- To compare engagement across employee segments
- To study tenure and role-based engagement patterns
- To analyze engagement versus attrition relationships
- To develop an interactive Power BI dashboard

TECHNOLOGIES USED

Microsoft Power BI – Dashboard development

Power Query – Data cleaning and transformation

DAX – KPI calculations

Microsoft Excel – Dataset management

DATA CLEANING AND PREPROCESSING

- Removed duplicate records
- Validated ordinal scales (1–4)
- Corrected data types
- Checked missing satisfaction values
- Standardized categorical labels
- Created CareerStage categories
- Created Distance categories
- Created normalized engagement components
- Built DAX measures and calculated columns

EXPLORATORY DATA ANALYSIS (EDA)

1. KPI Metrics

- Engagement Index
- Burnout Risk Score
- Work-Life Balance Index
- Satisfaction Stability Score
- Workload Stress Indicator
- Attrition Rate

2. Employee Engagement Analysis

3. Burnout Risk Analysis

4. Workload Stress Analysis

5. Career Stage Analysis

6. Engagement vs Attrition Analysis

BUSINESS INSIGHTS

- Employees with overtime and poor work-life balance demonstrated elevated burnout risk.
- Lower engagement scores were associated with employee attrition.
- Mid-career employees showed signs of declining engagement.
- Long-distance commuters experienced higher workload stress.
- Some departments displayed higher concentrations of burnout-prone employees.

BUSINESS RECOMMENDATIONS

- Introduce proactive employee wellness initiatives.
- Reduce excessive overtime workloads.
- Provide targeted support for high-risk employee groups.
- Conduct periodic employee engagement assessments.
- Implement role rotation programs.
- Improve work-life balance support mechanisms.

ADVANTAGES OF THE PROJECT

- Interactive and user-friendly dashboard
- Faster HR decision-making
- Early burnout detection
- Preventive employee intervention capability
- Dynamic workforce analysis

LIMITATIONS

- Dataset is static
- No real-time HR integration
- Burnout assumptions rely on available variables
- No predictive machine learning models

FUTURE SCOPE

- AI-based burnout prediction
- Employee attrition forecasting
- Machine learning risk classification
- Real-time HR database integration
- Cloud-based HR monitoring system

CONCLUSION

The Employee Engagement, Satisfaction, and Burnout Diagnostic Analysis project successfully provides a proactive HR analytics solution for Palo Alto Networks.

Using Power BI, Power Query, and DAX, the project transforms employee data into actionable insights related to engagement, burnout risk, employee satisfaction, and workforce well-being.

The dashboard supports early-warning diagnostics, enabling HR teams to identify engagement deterioration before employee attrition occurs.

REFERENCES

- Microsoft Power BI Documentation
- DAX Documentation
- Power Query Documentation
- HR Analytics Concepts
- Employee Engagement Research Literature
- Palo Alto Networks Employee Dataset

APPENDIX

Features Implemented

- KPI Cards
- Engagement Index
- Burnout Risk Dashboard

- Work-Life Balance Analysis
- Career Stage Analysis
- Attrition Analysis
- Manager Action Panel
- Department Filters
- Job Role Filters
- Overtime Toggle
- Interactive Slicers